# Team Details



**Name of Team**

NUCLEOFORGE

**Name of Members**

1. MANNAT THAKRAL
2. MOKSHITA POPLI
3. JITESH
4. PAAVAN



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# Problem Statement



## Problem statement and objective

THE PROBLEM

Students learn from scattered sources like PDFs, YouTube lectures, and GitHub repos every single week. Traditional storage apps lack semantic awareness; they save files but do not understand how concepts actually relate to each other. Because of this “forgetting curve,” students lose hours re-learning the exact same technical topics before hackathons and exams with zero visibility into their knowledge gaps.

| | |
|---|---|
| **Notes die in folders** | We dump notes into files and apps that store text but never think about it. |
| **Ideas stay disconnected** | Linking concepts is manual work, so hidden links between topics stay hidden. |
| **Nothing tests recall** | Students forget what they study. No tool checks retention on their own notes. |

OBJECTIVE

Build Human Cache: an AI-powered second brain that turns raw, unstructured notes into an interactive knowledge graph, then tests users on their own notes so learning actually sticks.

**The Objective:** To build an AI-powered “Second Brain” that dynamically models a user’s knowledge, explicitly maps relationships between scattered concepts, and predicts when a concept is about to be forgotten to trigger active recall.

Capture → Understand → Connect → Recall



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# Solutions



## Our solution

**WHAT IT IS** Human Cache is an AI-built, self-growing knowledge graph and predictive memory engine.

**HOW IT WORKS**

### 1 Ingest

Paste notes or add any source (PDFs, videos and code). No folders needed.

### 2 Extract

AI automatically extracts topics and entities, validated by Pydantic.

### 3 Connect

Semantic embeddings represent each concept as a vector, mathematically connecting related topics into a live knowledge graph.

### 4 Score

A weighted memory-decay model scores each concept's memory strength based on revision recency, practical usage and quiz accuracy.

**HOW IT SOLVES THE PROBLEM**

Instead of waiting for the user to forget, a recommendation layer ranks weak concepts by urgency and automatically generates short, targeted revision quizzes on demand, before the knowledge fades.



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# Tech Stack We Are Using...



| | |
|---|---|
| **Frontend**<br>The face of the app | Next.js + React + TypeScript, Tailwind CSS, React Flow (live on Vercel) |
| **Backend**<br>The brain of the app | Node.js with Express (and Python, FastAPI for AI processing), Pydantic (strict Node / Edge schemas) |
| **Database**<br>Permanent memory | PostgreSQL (Supabase / Neon) with SQLModel: Users, Notes, Nodes, Edges; pgvector for semantic vector storage |
| **Authentication**<br>User accounts | Firebase Auth / Clerk; per-user accounts via the Users table (auth provider to be finalized) |
| **Code Execution / APIs**<br>AI extraction layer | OpenAI Embeddings API; LLM API (OpenAI / Gemini / Claude) with Function Calling / JSON mode, also for generative quizzes |
| **Other tools**<br>Graph maths and hosting | NetworkX / React Flow (Knowledge Graph visualization; graph maths for quizzes), Vercel for deployment, Render / Railway |



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